FNAN 883, Homework 1
Testing Asset Pricing Models

1 Learning Objectives

- Understand and apply time-series and cross-sectional techniques for testing asset pricing models.
- Develop programming, presentation, and econometric skills that will be useful in your future research.
- Appreciate the empirical implications of theoretical asset pricing models and their success in explaining the cross-section of stock returns.

2 Goal

In this homework, you will test whether three asset pricing models explain cross-sectional differences in portfolio returns. The three asset pricing models are:

- **Asset Pricing Models**:
  - the CAPM (MKT factor) – a single-factor asset pricing model
  - the Fama and French (2015) 5-factor model (MKT, SMB, HML, RMW, and CMA) [aka FF5]
  - the Hou, Xue, and Zhang (2015) q-factor model (MKT, ME, IA, and ROE) [aka HXZ]
- **Tests**. To test all three models, you will use
  - the Gibbons, Ross, and Shanken (1989) [GRS] time-series test,
  - the Cross-Sectional Regression [CSR] test
  - the Fama and MacBeth (1973) method
- **Test Assets**. Your test assets will be the 32 portfolios formed based on triple sorts on size, book-to-market, and investments.

3 Data

The data are the monthly value-weighted returns of 32 portfolios formed on size, profitability, and investment (2×4×4). These are available on Prof. Kenneth French's Data Library website. The data are available from July 1963 to July 2026. The monthly returns of the Fama and French (2015) five factors are available on the Data Library website and on WRDS. The Hou, Xue, and Zhang (2015) q-factors are available on WRDS, and there is also a data

manual on the WRDS website: https://wrds-www.wharton.upenn.edu/documents/1104/q_factors_manual.pdf

The homework should include an appendix with your code and an explanation of what you are doing in the different sections of your code file.

4 Paper format

The homework should be written as a paper in the format and language of academic papers. This will help you practice writing a paper and 'promoting' your results. Try to write in an academic paper format, focusing on the main message, saying a lot with fewer words, editing and revising until you feel the write up is polished. This does not happen on the first draft. I expect the 'paper' to be about 3-5 pages.

1. *Abstract*: (give the 'punchline'=what is the biggest finding(s), stick to 100 words or less).

2. *Introduction*: Typically you would have an introduction, which briefly describes the literature/background, your contribution, why it's important, and the main results. The abstract and introduction are the selling point of your ideas and findings, so make sure they are strong, well written, to the point, give credit to all of your contributions, and stress the main message of your paper.

3. *Methodology*: describe the methodology used, using equations, motivating them (i.e. what does the equation measure), and giving proper credit to whomever came up with this methodology.

4. *Results*: describe and discuss your results. Make sure the reader can follow the descriptions and results in each table without looking at the tables. Describe as much as needed from the table, stressing the main results. You don't need to describe every single number in the tables, but give a few numbers that are important. Use tables and graphs (some ideas for tables and graphs are provided in the homework).

5. *Conclusion*: briefly describe what is the take-away from your paper, the main points the reader should remember. Stick to one paragraph for purposes of the homework.

6. *Tables*: The tables in working papers are included at the end of the paper, not within the text. Include each table on a separate page and make sure to have a description of what analysis you did to get the numbers in the table and what the numbers in the table represent (returns, test statistics, ...).

5 List of tables

5.1 Time series test, TABLE I

Use the Gibbons, Ross, and Shanken (1989) [GRS] method to test whether a factor or a set of factors is efficient:

1. Report the GRS test statistic, its p-value, and the 99th, 95th, and 90th percentile critical values for the test.

2. Test all three models: CAPM, FF5, and HXZ.

5.2 Cross-sectional tests

5.2.1 Cross-sectional regression, TABLE II

Use the cross sectional regression [CSR] method to test whether cross-sectional differences in returns are explained by differences in betas:

1. Report the estimated coefficients (lambdas) from the second-pass cross-sectional regression, along with their t-statistics (with and without the Shanken correction).

2. Report the Wald test statistic along with its p-value, as well as the 95% critical value.

3. Report results with and without intercept.

4. Test all three models: CAPM, FF5, and HXZ.

5.2.2 Fama-MacBeth cross-sectional regression, TABLE III

Use the Fama-MacBeth method to test whether cross-sectional differences in returns are explained by differences in betas:

1. Report the estimated coefficients (lambdas) from the second-pass cross-sectional regression, along with their sample t-statistics (with and without the Shanken correction). Include intercepts in both the time-series and cross-sectional regressions.

2. Report the average adjusted R-squared from the second-pass cross-sectional regressions (hint: you will have a time-series of R-squared for each month, report the average).

3. Test all three models: CAPM, FF5, and HXZ.

References

Fama, Eugene F., and Kenneth R. French, 2015, A Five-Factor Asset Pricing Model, *Journal of Financial Economics* 116, 1–22.

Fama, Eugene F., and James D. MacBeth, 1973, Risk, Return, and Equilibrium: Empirical Tests, *Journal of Political Economy* 81, 607–636.

Gibbons, Michael R., Stephen A. Ross, and Jay Shanken, 1989, A Test of the Efficiency of a Given Portfolio, *Econometrica* 57, 1121–1152.

Hou, Kewei, Chen Xue, and Lu Zhang, 2015, Digesting Anomalies: An Investment Approach, *Review of Financial Studies* 28, 650–705.